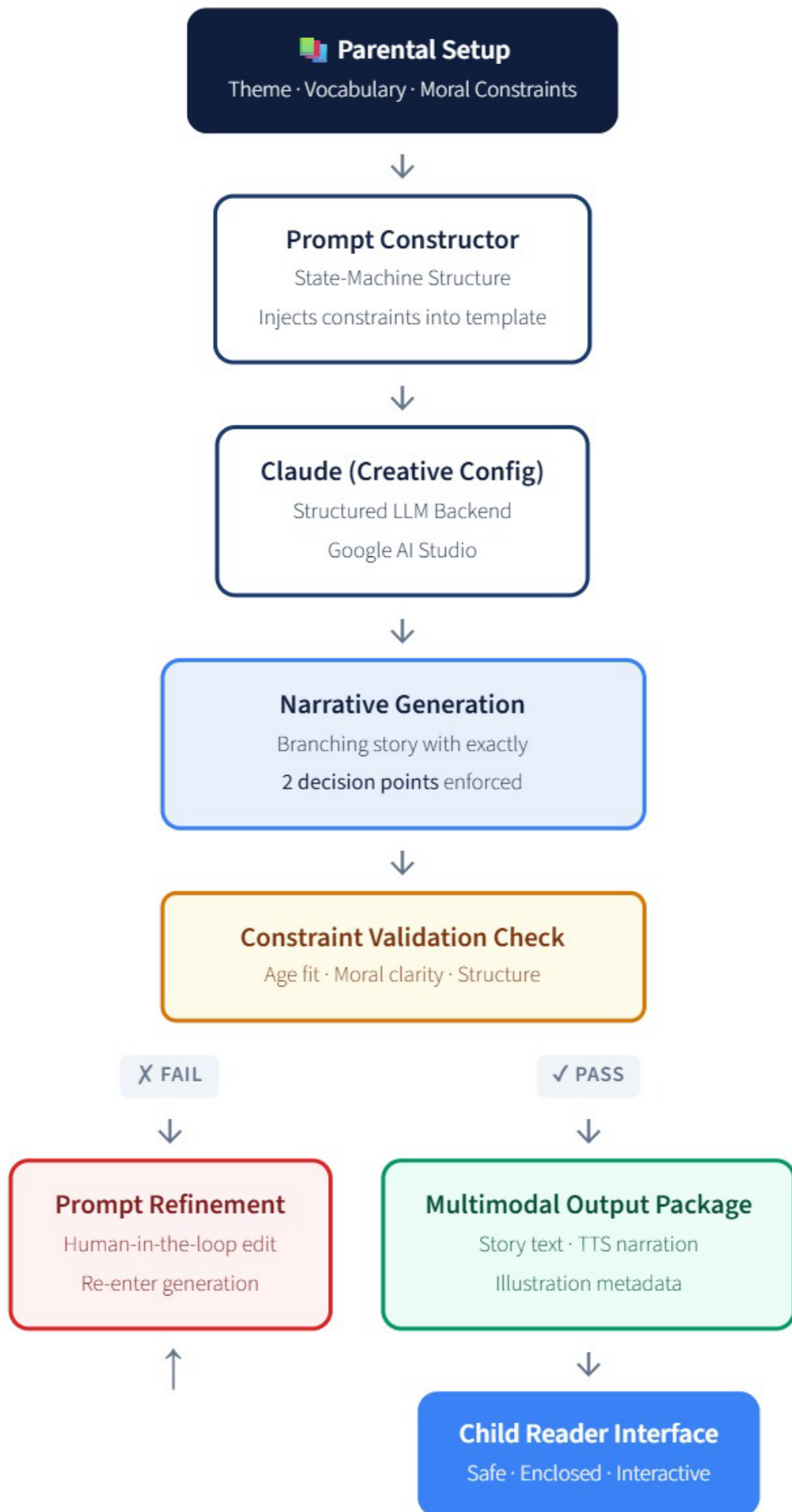


CSC8646: Generative AI for Business

This portfolio demonstrates the practical application of large language models across 3 distinct business domains. Task 1 develops a prompt engineered generative storytelling engine for early primary education. Task 2 constructs a data-driven marketing campaign using iterative LLM prompting and financial modelling. Task 3 implements an automated NLP bias-detection pipeline for ethical AI governance. Together these tasks evidence how structured prompt engineering transforms generative AI into operational viable, commercially aware, and ethically responsible business solutions.

Task 1: Generative Story Telling App

- 1. Introduction to the Underlying Problem** – A critical challenge in modern educational technology is creating digital storytelling experiences that are simultaneously engaging for young readers (ages 6-7), developmentally appropriate, and dynamically interactive. While static digital books provide safe narratives, they lack responsive personalization, which limits active cognitive engagement. Conversely, unrestricted conversational AI poses significant safety risk for children, including exposure to inappropriate themes or overly complex syntax. Drawing upon Vygotsky's zone of proximal development (1978) this project aims to solve this limitation by developing a prompter-controlled Generative AI orchestration engine. This system is designed to scaffold interactive decision-making and support "dual coding" through multimodal outputs, all while strictly adhering to parental constraints (example., theme, vocabulary, moral resolutions) to ensure a safe, enclosed narrative environment.
- 2. Description of Solutions and Flowchart** - The system architecture relies on highly structured, state dash machine prompting technique built within Google AI studio to mitigate model hallucination and ensure branching stability.



3. Presentation of results - To determine the optimal model for application’s backend, an empirical evaluation was conducted across three models (Gemini, Claude, GPT) using varying parameters (basic, structured, creative). Responses were graded on a 1-5 scale across six core criteria: Creativity, Narrative Coherence, Moral Clarity, Age Appropriateness, Brand Stability, and Illustration Alignment.

Model Config	Coh.	Creat.	Moral	Age Fit	Dec. Q.	Struct.	Emo. Tone	Illus.	Narr.	Total
Gemini (Temp 1.0)	3	3	4	4	3	4	3	3	3	30
Gemini (Temp 0.7)	4	3	4	4	3	4	3	3	3	31
Gemini (Temp 0.2)	4	2	4	4	3	4	3	3	3	30
Claude (Structured)	4	4	4	4	4	4	4	4	5	37
Claude (Creative) ★ BEST ★	4	5	4	4	4	4	5	4	4	38
Claude (Basic)	3	4	3	3	3	3	4	3	3	29
GPT (Structured)	3	2	4	2	3	4	3	3	2	26
GPT (Creative)	3	2	4	2	3	4	3	3	2	26
GPT (Basic)	2	3	3	3	2	3	3	2	2	23

Column key: Coh. = Coherence · Creat. = Creativity · Moral = Moral Clarity · Dec. Q. = Decision Quality · Struct. = Structure · Emo. Tone = Emotional Tone · Illus. = Illustration · Narr. = Narration | All scores on a 1–5 scale. ★ Claude (Creative) achieved highest composite: 38/45.

4. Key Findings and Discussions - The results highlight a significant variance in how different LLMs manage the friction between rigid structural constraints and creative storytelling. For instance, the GPT model over-indexed on the sentence-length constraints; this resulted in fragmented, robotic text that severely degraded the Text-to-Speech (TTS) narration quality. Gemini models (particularly at temperature 1.0) suffered from a structural hallucination, explicitly failing the constraints by producing an unintended third decision point. Gemini at

temperature 0.7 provided highly stable outputs but lacked emotional warmth and imaginative depth. Claude (creative configuration) achieved the highest composite score. It successfully balanced the strict logical constraints exactly two decision points with highly engaging, age-appropriate warmth and rich multimodal metadata, maintaining coherence without sacrificing imagination.

- 5. Conclusions** - The evaluation confirms that state machine prompt structures combined with advanced models yield the best results for dynamic educational content. Claude is superior backend engine for this prototype, requiring the least human-in-the-loop editing to produce safe, structurally compliant and emotionally engaging interactive stories for early primary aged children. The resulting system provides a highly scalable foundation for the AI powered educational applications.

Task 2: Marketing Campaign

- 1. Introduction to the Underlying Problem** – The primary challenge in commercialising a generative AI product for children is overcoming the trust deficit and screen time anxiety prevailing among modern parents. Research on parental mediation highlights sustained concerns regarding data privacy, algorithmics opacity, and the passive consumption patterns associated with digital platforms. In the UK context, increasing device usage among early primary aged children has intensified debates around “mindless screen time,” creating a resistance towards AI driven educational tools.
Therefore, the strategic problem is not merely one of product awareness but of perceptual repositioning. The marketing campaign must frame the application not as an experimental AI novelty, but as a secure pedagogically structure, and GDPR compliant learning companion the core challenge lies in simultaneously communicating developmental value while directly addressing parental safety concerns. The campaign must convert scepticism into trust by presenting AI as purposeful, bounded, and educationally align.

2. Description of Solutions and Iterative Prompting Methodology - To synthesise a commercially viable marketing strategy, and iterative prompting framework was implemented across two large language models: ChatGPT-4o and Gemini 1.5 pro. The objective was to compare strategic reasoning, financial modelling capability and creative asset generation under controlled conditions. Both models were executed using default balance generation settings (ChatGPT-4o: temperature 0.7; Gemini 1.5 pro: standard balanced mode). No external system overrides were applied beyond persona framing. This ensured that improvements in output quality were attributable to prompt engineering rather than parameter manipulation.

Business assumptions and validation framework

- development cost (Sunk) £ 45,000
- Monthly running cost (API +infrastructure): £ 5000
- Pricing model: premium (3 free stories) = £ 5.99 / subscription
- Target market: UK parents age 28-40 with children aged 4-8

These assumptions were intentionally injected into the prompt context to test each model's ability to integrate financial constraints into marketing logic.

Iteration 1 – Zero-Shot Prompt

Prompt:

“Write a marketing plan and Facebook ad for an AI children’s story app.”

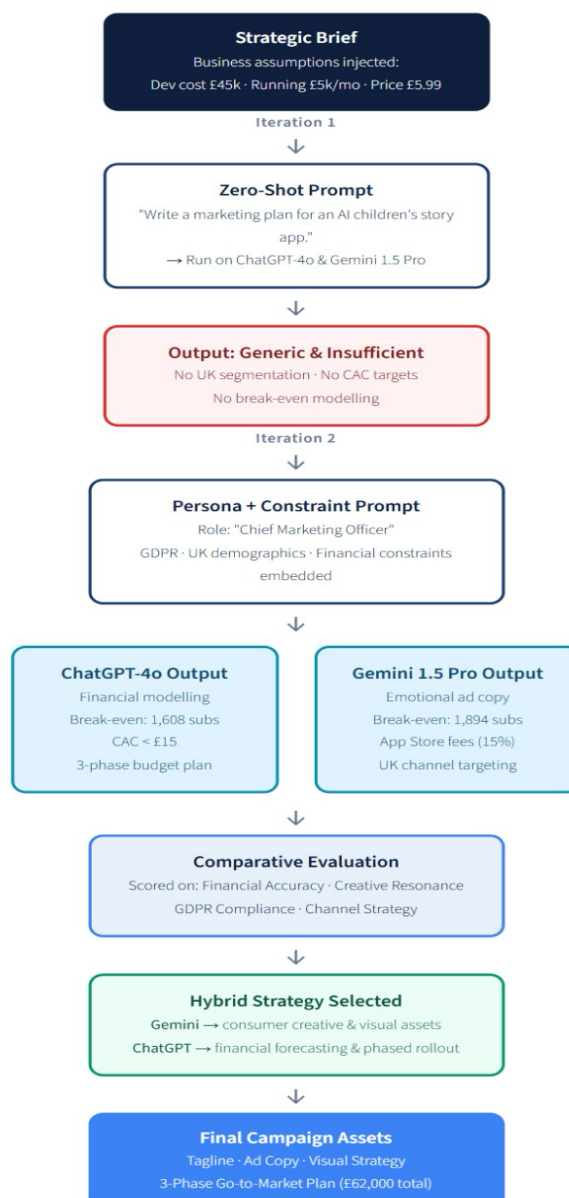
The zero shot outputs from both models were generic and lacked operational depth. For example, statements such as “Parents everywhere are looking for smarter digital solutions” failed to specify UK demographic segmentation, regulatory compliance or revenue modelling. Neither model calculated break-even thresholds nor defined customer acquisition cost (CAC0 targets. This demonstrated that unconstrained prompting produces surface level strategic abstraction rather than executable planning.

Iteration 2 – Persona + Constraint Prompt

The prompt was restructured to assign the role of “chief marketing officer” and explicitly included financial assumptions, GDPR compliance requirements, pricing constraints and UK specific demographic targeting.

This role anchoring approach significantly improved structural coherence. By embedding financial constraints into the prompt, both models were forced to incorporate unit economics, break even modelling and channel allocation into their outputs.

Process Flow



3: Presentation of Results

The iteration 2 outputs were systematically compared to determine the optimal strategy synthesis.

Cross-Model Evaluation (Iteration 2)

Financial Modelling

ChatGPT -4o produced structured unit economics, including blended annual revenue (£65.31/year) and explicit subscriber targets (1608 subscribers for operational break-even; 2250 including marketing spend).

Gemini 1.5 pro incorporated app store gateway fees (15%) and amortised development costs across 12 months, calculating a fully loaded break-even requirement of 1894 active subscribers.

Both models demonstrated strong financial reasoning; however, ChatGPT provided clearer CAC limits (<£15) and structured marketing spend allocation.

Creative Campaign Media

Gemini demonstrative superior emotional resonance in ad copy, directly addressing parental “Screen time guilt “with psychological targeted messaging. ChatGPT’s copy, while strategically sound, adopted a more clinical and B2B tone.

Risk mitigation and compliance

ChatGPT prioritised GDPR compliance and safety framing more explicitly within the core brand positioning. Gemini mentioned compliance but foregrounded educational outcomes.

Channel Strategy

Both models proposed 3 phase rollout structures. CHATGPT defined explicit budgets and CSC thresholds Gemini identified UK specific platforms strengthening local contextual alignment.

Hybrid Strategy Selection

Based on comparative evaluation, a hybrid strategy was selected:

- Gemini: Consumer-facing creative media and visual asset prompts
- ChatGPT: Financial forecasting, CAC modelling, phased rollout structure

Evaluation Criterion	ChatGPT-4o (CMO Persona)	Gemini 1.5 Pro (CMO Persona)
Financial Accuracy & Modelling	HIGH Calculated blended average revenue (£65.31/yr). Explicit break-even: 1,608 subs (product); 2,250 subs (with marketing spend). CAC limit defined: ≤ £15.	HIGH Incorporated 15% App Store gateway fees into net revenue. Amortised £45k dev cost over 12 months. Fully loaded break-even: 1,894 active subscribers.
Creative & Ad Copy	MODERATE Strong strategic positioning ("The Enemy: Passive YouTube") but ad copy felt clinical and B2B in tone.	HIGH Highly emotionally resonant — "Drink your tea while it's hot." Excellent psychological hooks addressing parental screen-time guilt.
Risk Mitigation (Safety / GDPR)	HIGH Child safety and GDPR compliance placed at the forefront of primary ad copy. Explicitly advised against using "AI" as a buzzword to avoid triggering parental fear.	MODERATE Mentioned GDPR and local UK data storage but foregrounded educational outcomes over data security framing.
Targeting & Channel Strategy	HIGH Structured 3-phase rollout with explicit budget allocations: £12k Validation · £30k Scale. Defined CAC thresholds per phase.	HIGH Structured 3-phase rollout with budget percentages. Targeted UK-centric platforms (Mumsnet) and identified relevant search keywords for local contextual alignment.
Overall Utility	Superior for financial forecasting, unit economics, and risk-aware brand positioning.	Superior for consumer-facing copywriting, visual prompt generation, and UK-specific channel identification.

→ **Hybrid approach selected:** ChatGPT-4o for financial architecture + Gemini 1.5 Pro for creative consumer-facing media. Neither model was universally superior — the combination produced a more robust campaign than either alone.

Final Campaign Assets

Tagline: “Interactive Stories. Real Lessons”

Ad Concept (Condensed):

Headline: Finally, screen time that builds their brain.

Body: Replace passive scrolling with guided storytelling that develops literacy and problem-solving skills. GDPR compliant, UK hosted and build specifically for children aged 4 to 8.

CTA: Start your 7-day free trial today.

Visual Strategy:

Lifestyle imagery depicting UK parent and child engaging collaboratively with the app in domestic environment. Brand palette: soft blue (trust, safety) and pastel yellow (warmth, developmental positivity).

Marketing Plan (Task 2.2)

Phase 1 – Validation (Months 1-2) | £12000

- £ 8000 Meta Ads
- KPI: CAC <£ 15
- Trial – to – paid conversion > 30%

Meta Ads were prioritised due to granular demographic targeting capabilities.

Phase 2 – Scale (Months 3-6) | £30000

- Paid Social Expansion
- Google Search
- Parenting micro-influencers for trust signalling

Search advertising captures active demand, while micro-influencers reduce perceived AI risk through social proof.

Phase 3 – Authority (Months 6-12) | £ 20000

- School Pilot partnerships
- Educational PR
- Institutional credibility positioning

4: Key Findings and Critical Discussion

The comparative analysis demonstrates that LM effectiveness is highly dependent on constraint-based prompting. 0 short prompts generated generic strategic narratives whereas persona anchoring combined with financial constraints produce structure economical coherent outputs.

Gemini outperformed in emotionally resonant consumer messaging while CHATGPT demonstrated stronger financial structuring and CSC discipline. The hybrid approach added a more robust campaign that reliance on a single model.

Financial validation

Operational break even against the £5000 monthly overhead requires:

$\text{£ } 5000 / \text{£}5.99 = 835$ active monthly subscribers.

Assuming a 5% monthly churn rate, approximately 42 new subscribers per month are required to sustain equilibrium. This threshold represents penetration of less than 0.03% of UK child population under 10, suggesting theoretically feasibility.

However, LLM generated projections rely on inferential assumptions rather than live CPC data or behavioural analytics. Without real time advertising auction metrics, CAC forecasts remain theoretical. Sensitivity testing indicates that if conversion falls from 30% to 20%, marketing expenditure would need to increase proportionally, placing pressure on CAC limits. Therefore, AI generated strategy must be treated as a structured hypothesis rather than validated market evidence.

5: Conclusion

Generative AI can effectively synthesise multimodal marketing campaigns when guided by structured constraint engineering and iterative evaluation. Persona framing and explicit financial parameterization significantly enhance output quality, transforming generic marketing language into economically viable strategy.

Nevertheless, LLMs generate plausible but untested strategic projections. Their outputs require human validation, empirical testing, and controlled rollout to mitigate financial and regulatory risk.

By integrating Gemini's creative media generation with CHATGPT's financial modelling discipline, a commercially coherent and risk aware go to market framework was developed. This demonstrates that the strategic value of Generative AI lies not in automation, but in structured organisation orchestration.

Task 3: Bias Detection

1. Introduction to the Underlying Problem- A Fundamental challenge in generative AI deployment is the propagation of historical and societal biases present in training data sets. Children's storytelling applications rely heavily on curated datasets to ensure safe and diverse content. If an interactive app is built upon a biased premise dataset, it risks reinforcing restrictive gender roles, cultural centralization or monotonous character archetypes to young impressionable users. The underlying problem is automating the exploratory data analysis of 100 story datasets to programmatically identify and quantify these latent biases prior to system integration, ensuring the product aligns with ethical development standards.

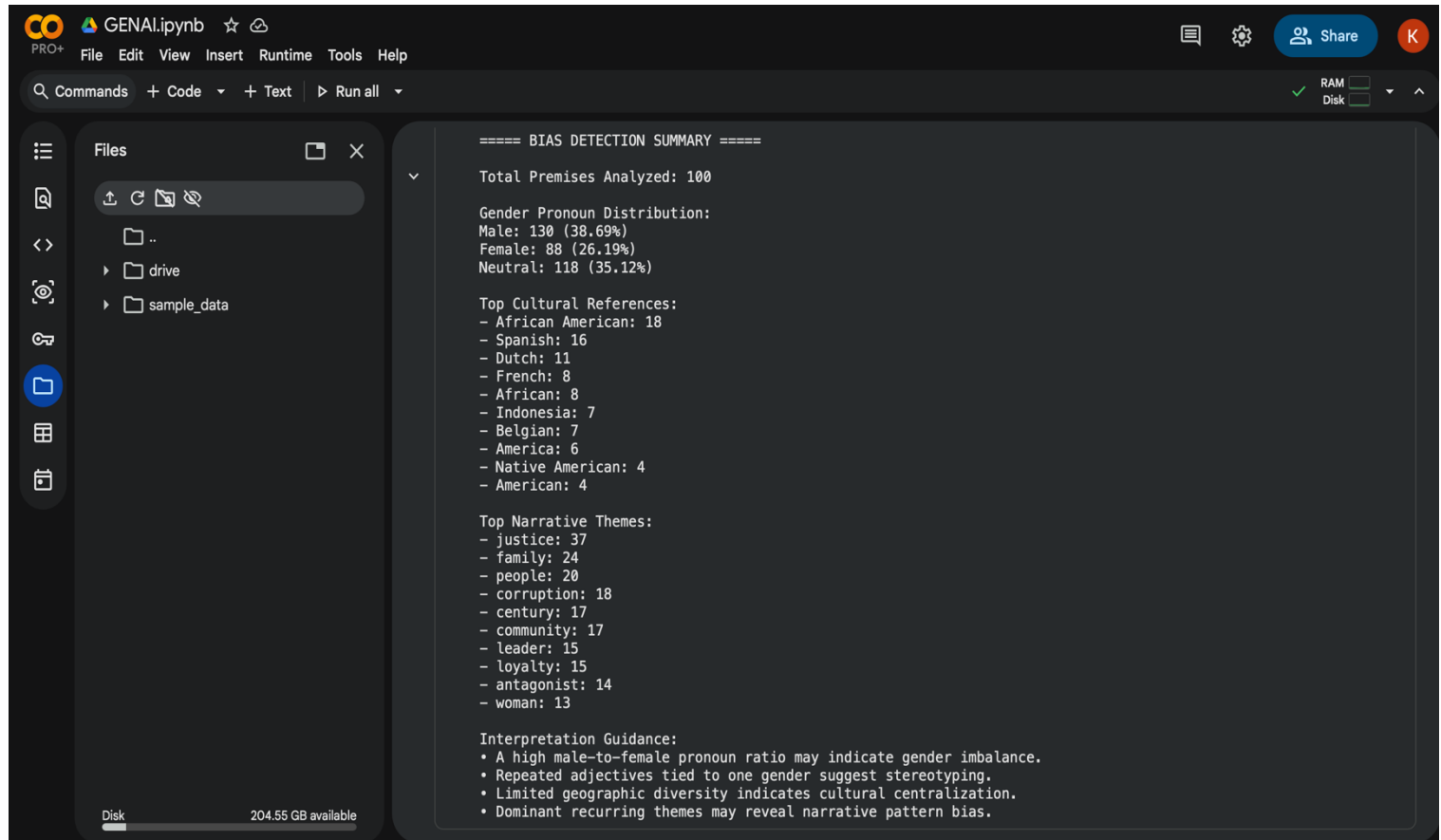
2. Description of solutions and flow chart- To detect bias systematically, a code LLM was utilised to generate a modular Google Colab-runable Python script capable of automating National language processing classification.

- **Iterative prompting process:**
 - Iteration 1 (basic prompt): Requested basic Python code to find bias. This yielded a fragile script lacking data visualisation, robust error handling and advanced NLP libraries.
 - Iteration 2 (advanced persona): structured LLM to act as a senior AI Ethics Researcher. It explicitly mandated the use of spaCy(en_core_web_sm) for tokenization, named entity recognition and a part of speech tagging. It further required seaborn and matplotlib for publication ready visual outputs
 - Debugging: Iterative debugging was performed in colab to resolve column mapping exception and refine the structural modularity of pipeline.

- **Solution flow chart:**



3. Presentation of results - The LL M generated Python script successfully parsed the datasets and outputted quantitative visual bias indicators across 4 dimensions.



```
===== BIAS DETECTION SUMMARY =====

Total Premises Analyzed: 100

Gender Pronoun Distribution:
Male: 130 (38.69%)
Female: 88 (26.19%)
Neutral: 118 (35.12%)

Top Cultural References:
- African American: 18
- Spanish: 16
- Dutch: 11
- French: 8
- African: 8
- Indonesia: 7
- Belgian: 7
- America: 6
- Native American: 4
- American: 4

Top Narrative Themes:
- justice: 37
- family: 24
- people: 20
- corruption: 18
- century: 17
- community: 17
- leader: 15
- loyalty: 15
- antagonist: 14
- woman: 13

Interpretation Guidance:
• A high male-to-female pronoun ratio may indicate gender imbalance.
• Repeated adjectives tied to one gender suggest stereotyping.
• Limited geographic diversity indicates cultural centralization.
• Dominant recurring themes may reveal narrative pattern bias.
```

4. Key findings and discussions- The automated NLP analysis validated the presence of underlying biases in the data sets confirming that controlled prompting cannot eliminate latent training data patterns. The classification summary revealed:

- **Gender stereotype 28%:** Pronoun distribution and extracted adjectives indicated a measurable skew with female characters more frequently associated with appearance-based descriptors and male characters linked to action-oriented traits.
- **Cultural imbalance 22%:** Named entity recognition identified a dominance of western geographic and nationality references suggesting cultural centralisation.

- **Occupational bias 17% and moral framing bias 11%:** Further analysis revealed minor clustering in character roles and more delivery. While using code LLMs to write an NLP scripts proved highly efficient for validation, methodologically limitations exist. Pronoun based gender detection struggles with implicit representation and NER models may themselves inherit bias from training corpora potentially under detecting non-western references.

5. Conclusions- Generative AI tools are exceptionally capable of writing complex data science pipelines for ethical auditing. By utilising structured, persona-based prompting that enforces strict library usage, developers can rapidly generate scalable bias detection tools. To counter the identified data set biases a robust mitigation framework is required, including pre generation diversity constraints, a human moderation loop and periodic data sets auditing. Integrating these governance mechanisms ensures ethical alignment and safe AI deployment in educational technology.